

(Integral Calculus)

Special functions

for $n > 0$ the Gamma (Γ) function is defined as

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$$

Note:-

$$\textcircled{1} \Gamma(1) = \int_0^{\infty} e^{-x} x^{1-1} dx$$

for $n=1$

$$= \int_0^{\infty} e^{-x} dx$$

$$\Gamma(1) = 1$$

$$\textcircled{2} \text{ for } n = \frac{1}{2}$$

$$\Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} e^{-x} x^{\frac{1}{2}-1} dx$$

$$= \int_0^{\infty} e^{-x} x^{-1/2} dx$$

$$\therefore \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Reduction Formula ($n > 0$)

$$\Gamma(n+1) = n \Gamma(n)$$

For positive integers

$$n=1 \Rightarrow \Gamma(1) = 1$$

$$n=2 \Rightarrow \Gamma(2) = \Gamma(1+1) = 1 \cdot \Gamma(1) = 1!$$

$$n=3 \Rightarrow \Gamma(3) = \Gamma(2+1) = 2 \cdot \Gamma(2)$$

$$= 2 \cdot 1! = 2!$$

$$n=4 \Rightarrow \Gamma(4) = \Gamma(3+1) = 3 \Gamma(3)$$

$$= 3 \cdot 2! = 3!$$

$$\boxed{\Gamma(n+1) = n!} \Rightarrow \text{only } n > 0$$

⑤ For $n < 0$ & n is non-integer

$$\boxed{\Gamma(n) = \frac{1}{n} \Gamma(n+1)}$$

* for $n=0$ the function is not defined

* Only true for the fractional values
not exists for negative integers.

$$* \Gamma(n) \cdot \Gamma(1-n) = \frac{\pi}{\sin n\pi}$$

→ For $n > 0$ and $m > 0$ the Beta(β) function is defined as

$$\boxed{\beta(m, n) = \int_0^1 (x)^{m-1} (1-x)^{n-1} dx}$$

Note:-

$$\boxed{\beta(m, n) = \beta(n, m)}$$

② By Taking $x = \sin^2 \theta$

$$\int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta = \frac{1}{2} \beta\left(\frac{m+1}{2}, \frac{n+1}{2}\right)$$

⑬ Relation b/w β & Γ functions

$$\beta(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)}$$

⑭ $\int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta = \frac{1}{2} \beta\left(\frac{m+1}{2}, \frac{n+1}{2}\right)$

$$= \frac{1}{2} \cdot \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{m+n+2}{2}\right)}$$

⑮ $\beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx$

⑯ $\beta(m, n) = \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx$

Session - 16/17

① Evaluate (i) $\Gamma(4.5)$

(ii) $\Gamma(-3.5)$

(iii) $\Gamma(7)$

(iv) $\Gamma\left(\frac{1}{4}\right) \times \Gamma\left(\frac{3}{4}\right)$

Sol:-

(i) $\Gamma(4.5) = \Gamma(3.5 + 1)$

$= 3.5 \Gamma(3.5) \Rightarrow 3.5 \Gamma(2.5 + 1)$

$= (3.5)(2.5) \Gamma(2.5)$

$$= (3.5)(2.5) \Gamma(1.5+1)$$

$$= (3.5)(2.5)(1.5) \Gamma(1.5)$$

$$= (3.5)(2.5)(1.5) \Gamma(1+0.5)$$

$$= (3.5)(2.5)(1.5)\left(\frac{1}{2}\right) \Gamma\left(\frac{1}{2}\right)$$

$$\left(\frac{3.5}{1.0}\right)\left(\frac{2.5}{1.0}\right)\left(\frac{1.5}{1.0}\right) \cdot \frac{1}{2} \sqrt{\pi}$$

$$\frac{105}{8} \times \frac{1}{2} \sqrt{\pi} = \frac{105}{16} \sqrt{\pi}$$

(ii)

$$\Gamma(-3.5) = \Gamma(n) = \frac{1}{n} \Gamma(n+1)$$

$$= -\frac{1}{3.5} \Gamma(-2.5)$$

$$= -\frac{1}{3.5} \cdot \frac{1}{-2.5} \Gamma(-2.5+1)$$

$$= -\frac{1}{3.5} \cdot \frac{1}{-2.5} \Gamma(-1.5)$$

$$= -\frac{1}{3.5} \cdot \frac{1}{-2.5} \cdot \frac{1}{-1.5} \Gamma(-1.5+1)$$

$$= -\frac{1}{3.5} \cdot \frac{1}{-2.5} \cdot \frac{1}{-1.5} \Gamma(-0.5)$$

$$= -\frac{1}{3.5} \cdot \frac{1}{-2.5} \cdot \frac{1}{-1.5} \cdot \frac{1}{-0.5} \Gamma(-0.5+1)$$

$$= \frac{1}{3.5} \cdot \frac{1}{2.5} \cdot \frac{1}{1.5} \cdot \frac{1}{0.5} \Gamma(0.5)$$

$$= \frac{1}{3.5} \cdot \frac{1}{2.5} \cdot \frac{1}{1.5} \cdot \frac{1}{0.5} \Gamma\left(\frac{1}{2}\right)$$

$$= \frac{1}{3.5} \cdot \frac{1}{2.5} \cdot \frac{1}{1.5} \cdot \frac{1}{0.5} \sqrt{\pi}$$

$$= \frac{10^2}{35} \cdot \frac{10^2}{25} \cdot \frac{10^2}{15} \cdot \frac{10^2}{5} \sqrt{\pi}$$

$$= \frac{16}{105} \sqrt{\pi}$$

(iii) $\Gamma(7)$

Sol:- $\Gamma(7) = 7 \cdot \Gamma(6)$

$$\Gamma(6+1) = 7 \cdot \Gamma(6)$$

$$\boxed{\Gamma(6+1) = 6!}$$

(iv) $\Gamma\left(\frac{1}{4}\right) \times \Gamma\left(\frac{3}{4}\right)$

$$\Gamma(n) \times \Gamma(1-n) = \Gamma\left(\frac{3}{4}\right) \times \Gamma\left(1-\frac{3}{4}\right)$$

$$= \Gamma\left(\frac{3}{4}\right) \times \Gamma\left(\frac{1}{4}\right)$$

$$= \frac{\pi}{\sin \pi n}$$

$$= \frac{\pi}{\sin \frac{3}{4}\pi}$$

//

③ Evaluate $\int_0^{\infty} x^5 e^{-x} dx$

Sol:-
 $= \int_0^{\infty} x^{6-1} e^{-x} dx$

$= \Gamma(6) = 5! = 120 //$

④ Evaluate $\int_0^1 \frac{x}{\sqrt{1-x^5}} dx$

Sol:- ~~$\int_0^1 x^{2-1}$~~ Take $x^5 = t$
 $x = t^{1/5}$

Differentiate w.r.t 't' on both sides.

$$\frac{dx}{dt} = \frac{1}{5} t^{-4/5} dt$$

Upper Limit $\Rightarrow x=1 \Rightarrow t=1$

Lower Limit $\Rightarrow x=0 \Rightarrow t=0$

Now, $\int_0^1 (t)^{1/5} \cdot \frac{1}{\sqrt{1-t}} \cdot \frac{1}{5} t^{-4/5} dt$

$= \frac{1}{5} \int_0^1 t^{-3/5} (1-t)^{-1/2} dt$

$= \frac{1}{5} \int_0^1 t^{(-3/5+1)-1} (1-t)^{1/2-1} dt$

$= \frac{1}{5} \int_0^1 t^{2/5-1} (1-t)^{1/2-1} dt$

$= \frac{1}{5} B\left(\frac{2}{5}, \frac{1}{2}\right) //$

$m = -\frac{3}{5} + 1$
 $m = \frac{2}{5}$

$n = -\frac{1}{2} + 1$
 $n = \frac{1}{2}$

$$= \frac{1}{5} \cdot \frac{\Gamma(\frac{2}{5}) \cdot \Gamma(\frac{1}{5})}{\Gamma(\frac{2}{5} + \frac{1}{5})} = \frac{1}{5} \cdot \frac{\Gamma(\frac{2}{5}) \sqrt{\pi}}{\Gamma(\frac{9}{10})}$$

⑤ Evaluate $\int_0^{\infty} \frac{x^4 (1+x^5)}{(1+x)^{15}} dx$

Sol: $\int_0^{\infty} \frac{x^4}{(1+x)^{15}} dx + \int_0^{\infty} \frac{x^9}{(1+x)^{15}} dx$

$m+1 = 4$
 $m = 5$
 $m+n = 15$
 $n = 15-5$
 $n = 10$

$$\int_0^{\infty} \frac{x^{5-1}}{(1+x)^{5+10}} dx + \int_0^{\infty} \frac{x^{10-1}}{(1+x)^{10+5}} dx$$

$$\Rightarrow B(5, 10) + B(10, 5)$$

$$= 2 B(5, 10)$$

$$= 2 \cdot \frac{\Gamma(5) \Gamma(10)}{\Gamma(5+10)} = 2 \times \frac{\Gamma(5) \Gamma(10)}{\Gamma(15)}$$

$$= 2 \times \frac{4! \times 9!}{14!}$$

⑥ $\int_0^{\pi/2} \sin^5 \theta \cos^{7/2} \theta d\theta = \frac{1}{2} B\left(\frac{m+1}{2}, \frac{n+1}{2}\right)$

$$= \frac{1}{2} B\left(\frac{6}{2}, \frac{9/2}{2}\right)$$

$$= \frac{1}{2} B\left(3, \frac{9}{4}\right) = \frac{1}{2} \cdot \frac{\Gamma(3) \Gamma(\frac{9}{4})}{\Gamma(3+\frac{9}{4})} = \frac{1}{2} \cdot \frac{\Gamma(3) \Gamma(\frac{9}{4})}{\Gamma(\frac{21}{4})}$$

$$\begin{aligned}
 f &= \frac{1}{2} \cdot \frac{\Gamma(\frac{9}{4})}{\Gamma(\frac{21}{4})} = \frac{\Gamma(\frac{9}{4})}{\Gamma(\frac{21}{4})} \\
 &= \frac{\Gamma(\frac{9}{4})}{\Gamma(\frac{17}{4}+1)} = \frac{\Gamma(\frac{9}{4})}{\frac{17}{4} \Gamma(\frac{17}{4})} \\
 f &= \frac{\Gamma(\frac{9}{4})}{\Gamma(\frac{9}{4})} = \frac{\Gamma(\frac{9}{4})}{\Gamma(\frac{9}{4})} \\
 &= \frac{\frac{17}{4} \Gamma(\frac{13}{4}+1)}{\frac{17}{4} \times \frac{13}{4} \Gamma(\frac{13}{4})} \\
 &= \frac{\Gamma(\frac{9}{4})}{\Gamma(\frac{9}{4})} = \frac{\Gamma(\frac{9}{4})}{\Gamma(\frac{9}{4})} \\
 &= \frac{\frac{17}{4} \times \frac{13}{4} \Gamma(\frac{9}{4}+1)}{\frac{17}{4} \times \frac{13}{4} \times \frac{9}{4} \Gamma(\frac{9}{4})} \\
 &= \frac{1}{\frac{17 \times 13 \times 9}{64}} = \frac{64}{17 \times 13 \times 9}
 \end{aligned}$$

Evaluate

① $\int_0^{\pi/2} \sqrt{\cos \theta} d\theta$

② $\int_0^3 \frac{1}{\sqrt{9-x^2}} dx$ x = 3t

m = 0

~~$\Gamma = 1/2$~~

~~$= \frac{1}{2} B\left(\frac{0+1}{2}, \frac{1/2+1}{2}\right) = \frac{1}{2} B\left(\frac{1}{2}, \frac{3}{2}\right)$~~

$\int_0^{\pi/2} \sqrt{\frac{\cos \theta}{\sin \theta}} d\theta = \int_0^{\pi/2} \cos^{1/2} \theta \sin^{-1/2} \theta d\theta$

$= \frac{1}{2} B\left(\frac{-1/2+1}{2}, \frac{1/2+1}{2}\right)$

$= \frac{1}{2} B\left(\frac{1/2}{2}, \frac{3/2}{2}\right) = \frac{1}{2} B\left(\frac{1}{4}, \frac{3}{4}\right)$

$$= \frac{1}{2} \cdot \frac{\Gamma(\frac{1}{4}) \cdot \Gamma(\frac{3}{4})}{\Gamma(\frac{1}{4} + \frac{3}{4})} = \frac{1}{2} \cdot \frac{\Gamma(\frac{1}{4}) \cdot \Gamma(\frac{3}{4})}{\Gamma(1)}$$

$$= \frac{1}{2} \cdot \frac{\sqrt{2\pi}}{1} = \frac{1}{\sqrt{2}} \cdot \frac{\pi}{1} = \frac{\pi}{\sqrt{2}}$$

$$\textcircled{3} \int_0^1 x^4 (1-x)^3 dx$$

06-03-25

$$\textcircled{2} \int_0^3 \frac{1}{\sqrt{9-x^2}} dx$$

Sol:- $x^2 = 9t$
 $x = 3t^{1/2}$

$$dx = 3 \left(\frac{1}{2} \right) t^{-1/2} dt$$

$$dx = \frac{3}{2} t^{-1/2} dt$$

U.L = $x = 3 \Rightarrow 9 = 9t \Rightarrow t = 1$

L.L = $x = 0 \Rightarrow 0 = 9t \Rightarrow t = 0$

$$= \int_0^3 \frac{1}{\sqrt{9-x^2}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{9-9t}} \cdot \frac{3}{2} t^{-1/2} dt$$

$$= \frac{3}{2} \int_0^1 \frac{1}{\sqrt{1-t}} t^{-1/2} dt$$

$$= \frac{1}{2} \int_0^1 \frac{1}{\sqrt{1-t}} t^{-1/2} dt$$

$$= \frac{1}{2} \int_0^1 (1-t)^{-1/2} t^{-1/2} dt$$

$$= \frac{1}{2} \int_0^1 (1-t)^{-1/2} t^{-1/2} dt$$

$$= \frac{1}{2} \int_0^1 t^{1/2-1} (1-t)^{1/2-1} dt$$

$$= \frac{1}{2} B\left(\frac{1}{2}, \frac{1}{2}\right)$$

$$= \frac{1}{2} \frac{\Gamma\left(\frac{1}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left[\frac{1}{2} + \frac{1}{2}\right]}$$

$$= \frac{1}{2} \frac{\sqrt{\pi} \cdot \sqrt{\pi}}{1}$$

$$= \frac{\pi}{2}$$

f Double Integral

$$\textcircled{1} \int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\textcircled{2} \int a f(x) dx = a \int f(x) dx + C$$

$$\textcircled{3} \int \sin(ax+b) dx = \frac{-\cos(ax+b)}{a} + C$$

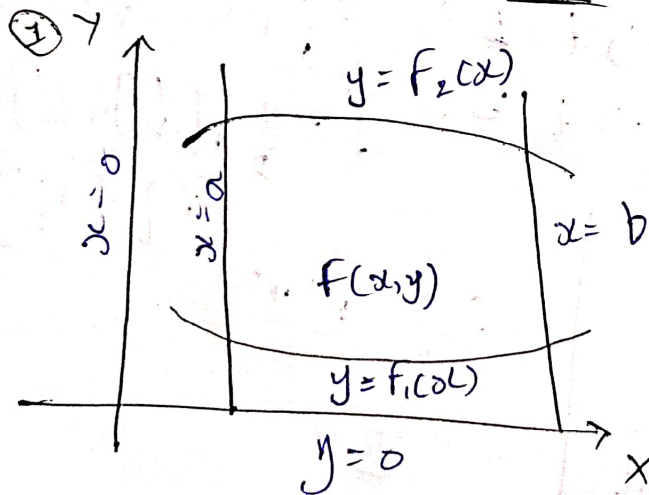
$$\textcircled{4} \int \cos(ax+b) dx = \frac{-\sin(ax+b)}{a} + C$$

$$\textcircled{5} \int \frac{1}{x} dx = \log x + C$$

$$\textcircled{6} \int \log x dx = x \log x + C$$

$$\textcircled{7} \int 1 dx = x + C$$

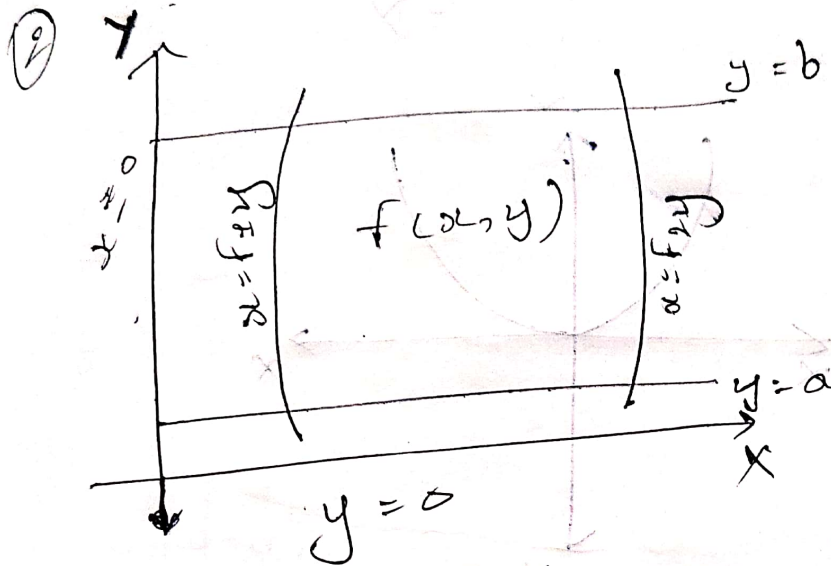
Evaluation of Double Integral



* First we take inner integration of $f(x,y)$.

$$\int_{x=a}^b \int_{y=f_1(x)}^{y=f_2(x)} f(x,y) dy dx$$

* First we take $f(x, y)$ w.r.t 'y' by treating 'x' as constant and then we take outer integration w.r.t 'x' in between $x=a$ and $x=b$.

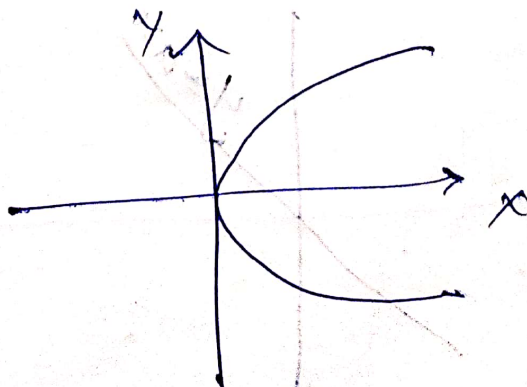


$$\int_{y=a}^b \int_{x=f_1(y)}^{x=f_2(y)} f(x, y) dx dy$$

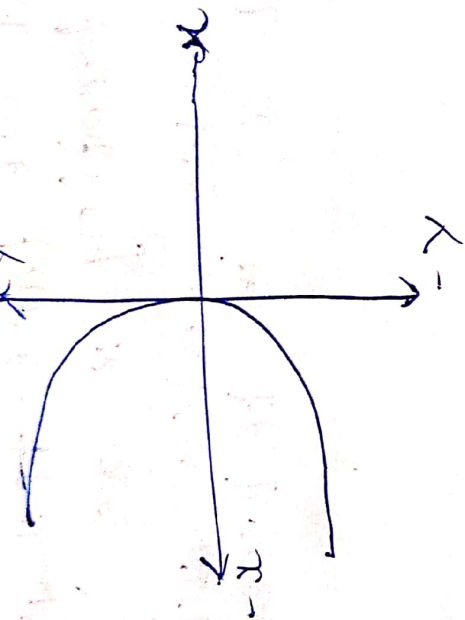
* First we take $f(x, y)$ and perform inner integration w.r.t 'x' by treating 'y' as constant and then we take outer integration w.r.t 'y' in between $y=a$ and $y=b$.

Note:-

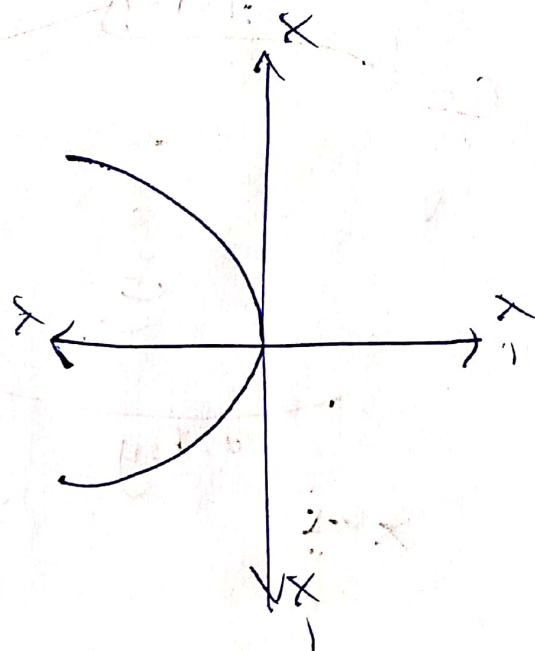
$$y^2 = kx \Rightarrow$$



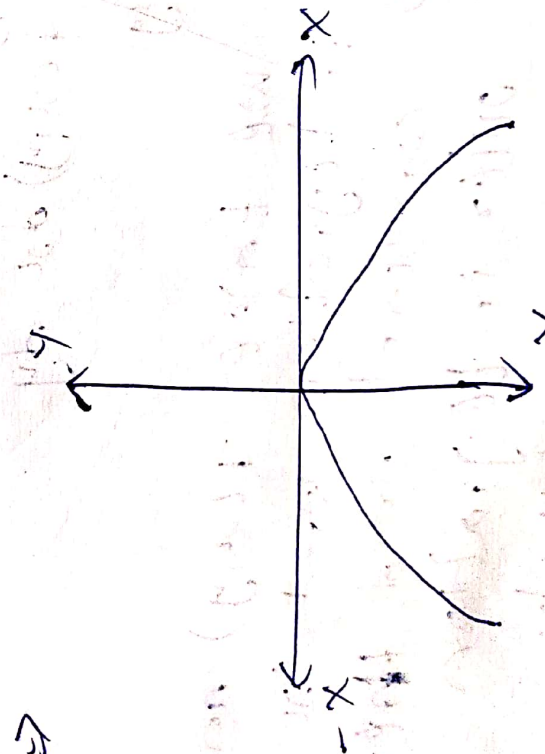
$$y^2 = -4ax \Rightarrow$$



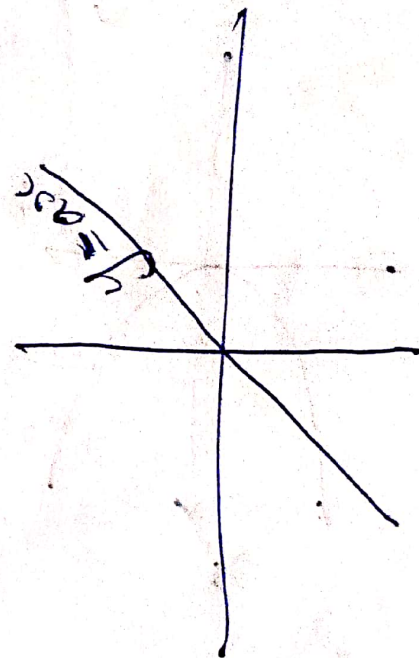
$$x^2 = 4ay \Rightarrow$$



$$x^2 = -4ay \Rightarrow$$

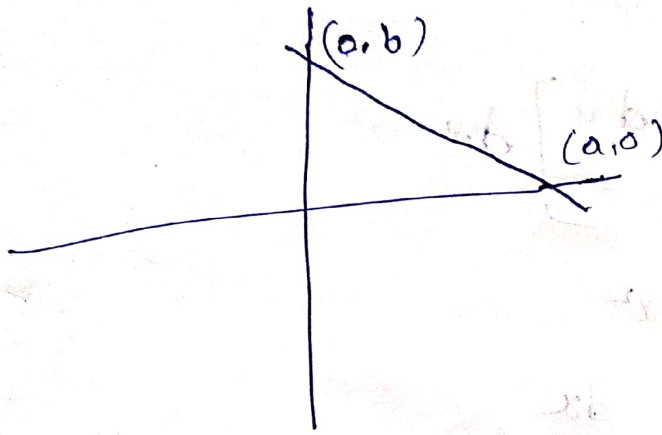


→ straight line passing through origin.



Intercept form

$$\frac{x}{a} + \frac{y}{b} = 1$$



Session - 18

① Evaluate $\int_0^1 \int_1^2 (x+y) dy dx$

$$\text{Sol:} = \int_0^1 \left[\int_1^2 (x+y) dy \right] dx$$

$$= \int_0^1 \left[\left(xy + \frac{y^2}{2} \right) \right]_{y=1}^2 dx$$

$$= \int_0^1 \left[\left(2x + \frac{4}{2} \right) - \left(x + \frac{1}{2} \right) \right] dx$$

$$= \int_0^1 \left[x + \frac{3}{2} \right] dx =$$

$$= \left[\frac{x^2}{2} + \frac{3}{2} x \right]_{x=0}^1$$

$$= \left(\frac{1}{2} + \frac{3}{2} \right) - \left[\frac{0}{2} + \frac{3}{2} \cdot 0 \right] = \frac{4}{2} = 2 //$$

(2) Evaluate $\int_0^4 \int_0^{\sqrt{4-x^2}} (xy) dx dy$

Sol:-

$$\int_{x=0}^4 \left[\int_{y=0}^{\sqrt{4-x^2}} (xy) dy \right] dx$$

$$\int_0^4 \left[x \frac{y^2}{2} \right]_0^{\sqrt{4-x^2}} dx$$

$$\frac{1}{2} \int_0^4 \left[(x \cdot (4-x^2)) - x \frac{(0)^2}{2} \right] dx$$

$$\frac{1}{2} \int_0^4 (4x - x^3) dx$$

$$\frac{1}{2} \int_0^4 \left[\frac{4x^2}{2} - \frac{x^4}{4} \right] dx$$

$$\frac{1}{2} \left[\frac{4(4)^2}{2} - \frac{(4)^4}{4} \right] - 0$$

$$\frac{1}{2} [32 - 64]$$

$$\frac{1}{2} [-32]$$

$$= -16$$

③ Evaluate $\int_0^1 \int_x^{x^2} (1+x) dy dx$

Sol:- $\int_{x=0}^1 \left[\int_{y=x}^{y=x^2} (1+x) dy \right] dx$

$= \int_{x=0}^1 [y + xy]_{x}^{x^2} dx$

$= \int_{x=0}^1 [x^2 + x^3] - [x + x^2] dx$

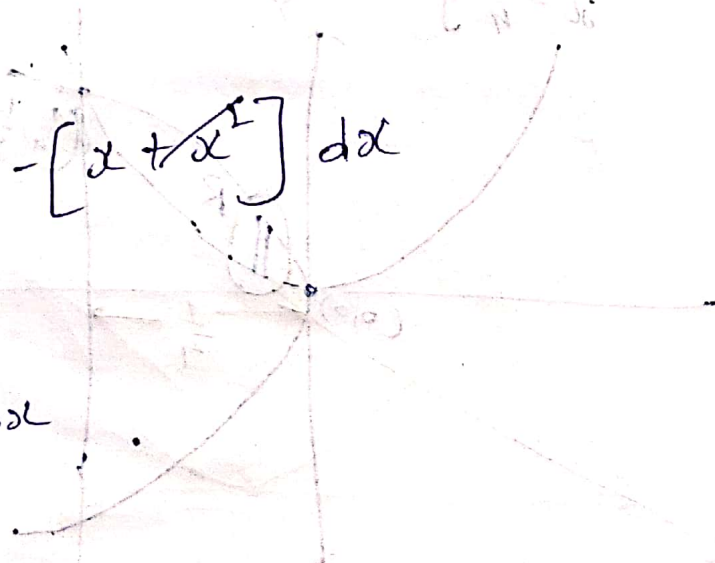
$= \int_0^1 (x^3 - x) dx$

$= \int_0^1 \left[\frac{x^4}{4} - \frac{x^2}{2} \right] dx$

$= \left[\frac{x^4}{4} - \frac{x^2}{2} \right] - [0] = \frac{1}{4} - \frac{1}{2}$

$= \frac{1}{4} - \frac{1}{2}$

$= \frac{1-2}{4} = -\frac{1}{4} //$

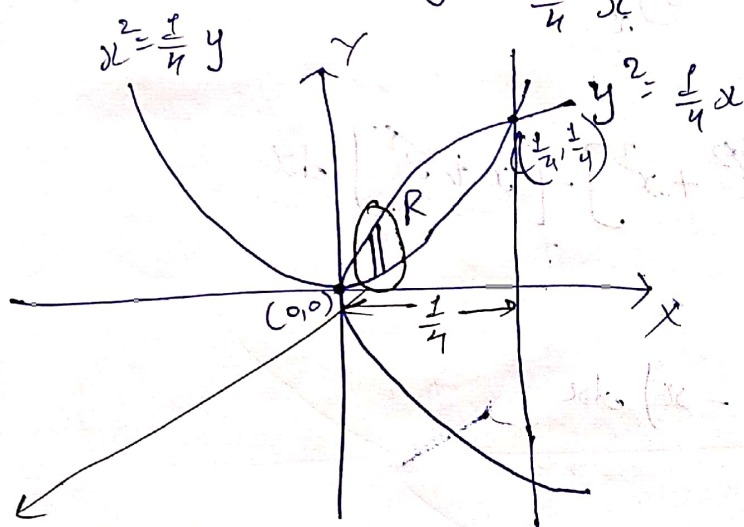


Q) Evaluate area of the region bounded by the parabolas $y = 4x^2$ & $x = 4y^2$

Sol:- Area = $\iint_R dx dy = \int \int dA$

R. $y = 4x^2 \Rightarrow x^2 = \frac{1}{4}y$

$x = 4y^2 \Rightarrow y^2 = \frac{1}{4}x$



Vertical strip.

* For vertical strip x^2 is lower bound & y^2 is upper bound.

So, $x^2 = \frac{1}{4}y \Rightarrow y = 4x^2$, $y^2 = \frac{1}{4}x$

$y = \frac{\sqrt{x}}{2}$

* For 'x' limits take from horizontally left to right.

$0 \rightarrow \frac{1}{4}$

$y^2 = 16x^4 \Rightarrow \frac{y}{4} = 16x^4 \Rightarrow 64x^4 - y = 0$

$x[64x^3 - 1] = 0$
 $x = 0 \mid x^3 = \frac{1}{64} \Rightarrow$

$\rightarrow x = 0 \Rightarrow y = 0 \Rightarrow (0,0)$

$\rightarrow x = \frac{1}{4} \Rightarrow y = \sqrt[3]{\frac{1}{64}} = \frac{1}{4}$

$A = \iint_R dx dy$

$= \int_{x=0}^{\frac{1}{4}} \int_{y=4x^2}^{\frac{\sqrt{x}}{2}} \left[\frac{1}{4} dy \right] dx$

$= \int_0^{\frac{1}{4}} \left(y \right)_{\frac{1}{4}x^2}^{\frac{\sqrt{x}}{2}} dx$

$= \int_0^{\frac{1}{4}} \left[\frac{\sqrt{x}}{2} - \frac{1}{4}x^2 \right] dx$

$= \left[\frac{1}{2} \cdot \frac{x^{3/2}}{3/2} - \frac{1}{4} \cdot \frac{x^3}{3} \right]_0^{\frac{1}{4}}$

$= \left[\frac{1}{2} \cdot \frac{2}{3} x^{3/2} - \frac{1}{12} x^3 \right]_0^{\frac{1}{4}}$

$= \left[\frac{1}{3} x^{3/2} - \frac{1}{12} x^3 \right]_0^{\frac{1}{4}}$

$= \left[\frac{1}{3} \left(\frac{1}{4} \right)^{3/2} - \frac{1}{12} \left(\frac{1}{4} \right)^3 \right]$

$$x[64x^3 - 1] = 0$$

$$x=0 \mid x^3 = \frac{1}{64} \Rightarrow x = \frac{1}{4}$$

$$\rightarrow x=0 \Rightarrow y=0 \Rightarrow (0,0)$$

$$\rightarrow x = \frac{1}{4} \Rightarrow y = \sqrt[4]{\frac{1}{64}} = \frac{1}{4} \Rightarrow \left(\frac{1}{4}, \frac{1}{4}\right)$$

$$A = \iint_R dx dy$$

$$= \int_{x=0}^{\frac{1}{4}} \int_{y=4x^2}^{\frac{\sqrt{x}}{2}} [1] dy dx$$

$$= \int_0^{\frac{1}{4}} \left(\frac{\sqrt{x}}{2} - 4x^2 \right) dx$$

$$= \int_0^{\frac{1}{4}} \left[\frac{\sqrt{x}}{2} - 4x^2 \right] dx$$

$$= \left[\frac{1}{2} \cdot \frac{x^{3/2}}{3/2} - \frac{4}{3} x^3 \right]_0^{\frac{1}{4}}$$

$$= \left[\frac{1}{2} \cdot \frac{2}{3} x^{3/2} - \frac{4}{3} x^3 \right]_0^{\frac{1}{4}}$$

$$= \left[\frac{1}{3} x^{3/2} - \frac{4}{3} x^3 \right]_0^{\frac{1}{4}}$$

$$= \left[\frac{1}{3} \left(\frac{1}{4}\right)^{3/2} - \frac{4}{3} \left(\frac{1}{4}\right)^3 \right] - 0$$

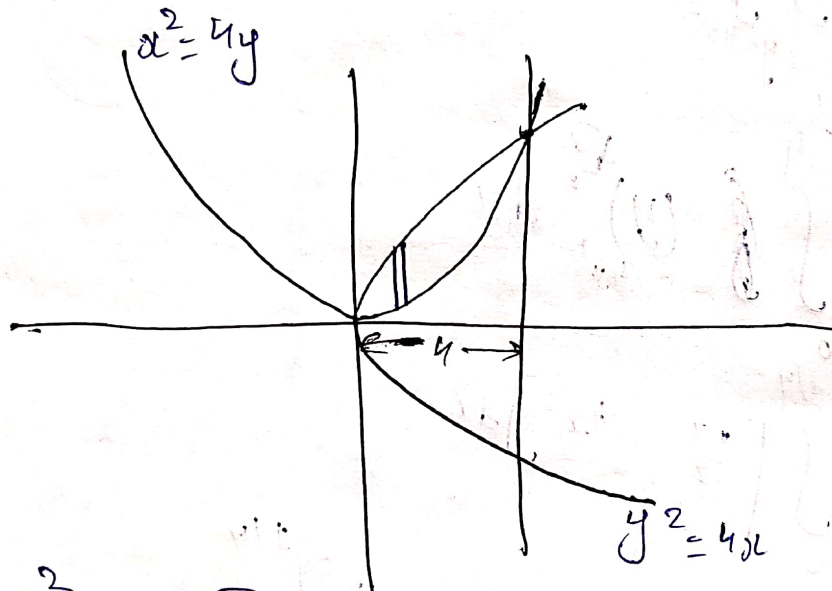
$$= \frac{1}{3} \cdot \left(\frac{1}{2}\right)^{3/2} - \frac{1}{3} \cdot \frac{1}{2}$$

$$= \frac{1}{3} \cdot \frac{1}{8} - \frac{1}{6}$$

$$= \frac{1}{24} - \frac{1}{48} = \frac{2-1}{48} = \frac{1}{48} //$$

⑤ Evaluate Area enclosed by parabolas
 $y^2 = 4x$, $x^2 = 4y$

$$A = \iint_R dx dy \int dA$$



$$* y: \frac{x^2}{4} \rightarrow 2\sqrt{x}$$

$$* x: 0 \rightarrow 4$$

$$y^2 = 4x$$

$$x^2 = 4y = x^4 = 16y^2$$

$$x^4 = 16 \cdot 4x$$

$$x^4 = 64x$$

$$x[x^3 - 64] = 0$$

$$x = 0, x = \sqrt[3]{64}$$

$$x = 0, x = 4$$

$$\downarrow$$

$$x = 0 \Rightarrow y = 0 (0,0)$$

$$x = 4 \Rightarrow y = 4 (4,4)$$

$$A = \int_{x=0}^4 \int_{y=\frac{x^2}{4}}^{\sqrt{x}} [dy] dx$$

$$A = \int_0^4 \left(y \right)_{\frac{x^2}{4}}^{2\sqrt{x}} dx$$

$$A = \int_0^4 \left(2\sqrt{x} - \frac{x^2}{4} \right) dx$$

$$A = \left[2 \cdot \frac{x^{3/2}}{3/2} - \frac{1}{4} \cdot \frac{x^3}{3} \right]_0^4$$

$$A = \left[2 \cdot \frac{2}{3} \cdot x^{3/2} - \frac{x^3}{12} \right]_0^4$$

$$A = \left[\frac{4}{3} x^{3/2} - \frac{x^3}{12} \right]_0^4$$

$$= \left(\frac{4}{3} (4)^{3/2} - \frac{4^3}{12} \right) - 0$$

$$= \frac{4}{3} \times 8 - \frac{64}{12}$$

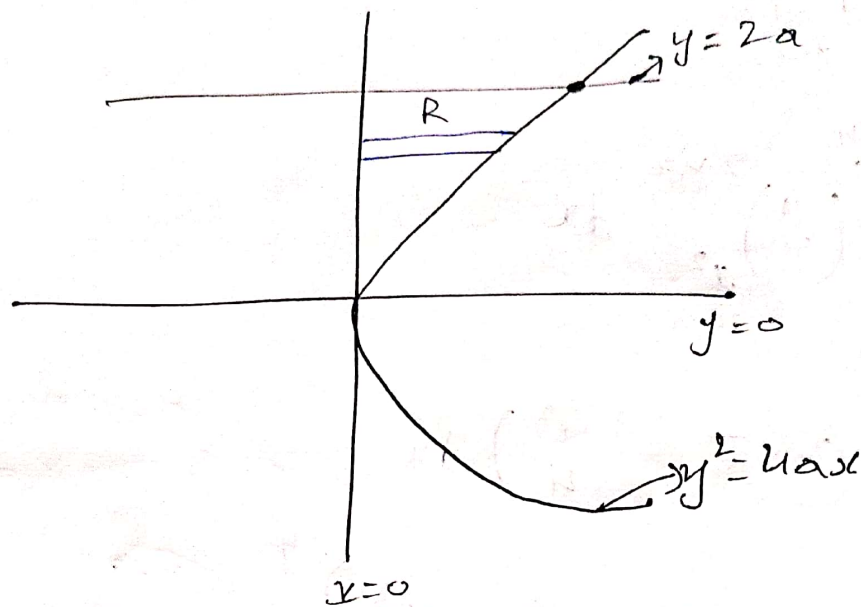
$$= \frac{32}{3} - \frac{16}{3} = \frac{16}{3} //$$

⑥ Evaluate $\iint_R xy \, dx \, dy$ where R is region bounded by y -axis, ordinate $y=2a$ and the curve $y^2=4ax$.

Sol:-

$$I = \iint_R xy \, dx \, dy$$

$$\begin{array}{l} y\text{-axis} \Rightarrow x=0 \\ y=2a \end{array} \quad \left| \quad y^2=4ax \right.$$



$$x: 0 \rightarrow \frac{y^2}{4a}$$

$y: 0 \rightarrow 2a$ Here take the horizontal strip, while finding 'x' & 'y' limits.

* So, 'x' starts from 0 to the upper bound which lies on the parabola so, $x = \frac{y^2}{4a}$

* For 'y' move vertically from 0 to the straight line $2a$.

$$I = \iint xy \, dx \, dy$$

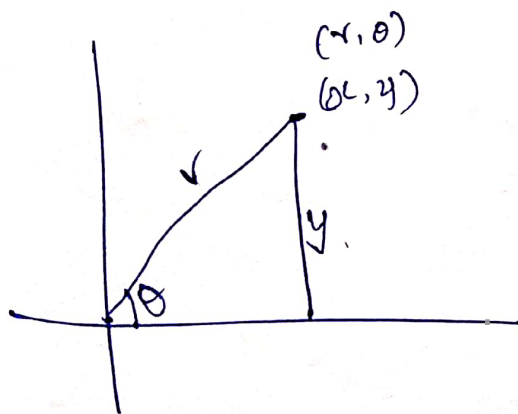
$$I = \int_{y=0}^{2a} \int_{x=0}^{y^2/4a} [(y \cdot x) \, dx] \, dy$$

$$I = \int_0^{2a} \left(y \cdot \left(\frac{x^2}{2} \right) \right)_0^{\frac{y^2}{4a}} \, dy$$

$$I = \int_0^{2a} y \cdot \left(\frac{\frac{y^4}{16a^2}}{2} \right) \, dy$$

$$\begin{aligned}
 I &= \frac{1}{2} \cdot \int_0^{2a} y \cdot \frac{y^4}{16a^2} dy \\
 &= \frac{1}{2 \cdot 16a^2} \int_0^{2a} (y^5) dy \\
 &= \frac{1}{32a^2} \left[\frac{y^6}{6} \right]_0^{2a} = \frac{1}{32a^2} \cdot \frac{(2a)^6}{6} \\
 &= \frac{1}{32a^2} \cdot \frac{2^6 \cdot a^6}{6} = \frac{2^6 \cdot a^6}{32 \cdot 6 \cdot a^2} = \frac{2^6 \cdot a^4}{192} = \frac{2^4 \cdot a^4}{24} = \frac{2^3 \cdot a^4}{3} = \frac{8a^4}{3}
 \end{aligned}$$

Polar Co-ordinates



$$\begin{aligned}
 \sin \theta &= \frac{y}{r} \Rightarrow y = r \sin \theta \\
 \cos \theta &= \frac{x}{r} \Rightarrow x = r \cos \theta \\
 x^2 + y^2 &= r^2 \cos^2 \theta + r^2 \sin^2 \theta \\
 x^2 + y^2 &= r^2 (\cos^2 \theta + \sin^2 \theta) \\
 x^2 + y^2 &= r^2 (1) \\
 \boxed{x^2 + y^2} &= \boxed{r^2}
 \end{aligned}$$

* For transforming cartesian to polar form.

$$\iint (x^2 + y^2) (dx dy)$$

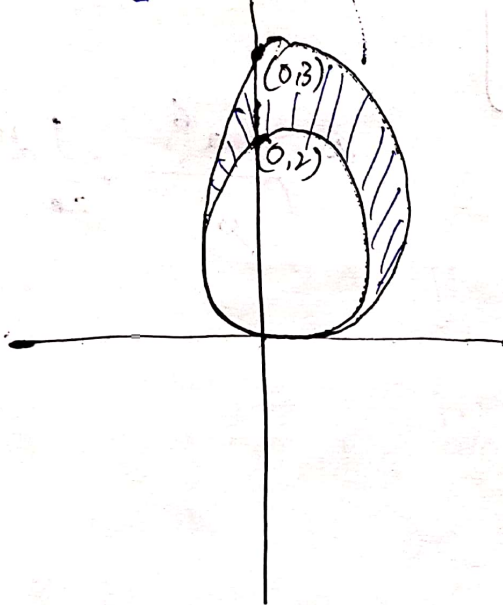
* $dx dy = r dr d\theta$ (substitution)
 $x = r \cos \theta, y = r \sin \theta$

① Evaluate $\int \int r^3 dr d\theta$ over the area included between the circles $r = 4 \sin \theta$ and $r = 6 \sin \theta$

Sol:- $I = \int \int r^3 dr d\theta$

$r: 4 \sin \theta \rightarrow 6 \sin \theta$

$\theta: 0 \rightarrow \pi$



* For the graph convert Cartesian into polar

~~$x = r \cos \theta$~~
 $\cos \theta = \frac{x}{r}$

$y = r \sin \theta$
 $\sin \theta = \frac{y}{r}$

$r^2 = 4y = 0 \Rightarrow x^2 + y^2 - 4y = 0$
 is circle with centre $(0, 2)$

$r = 6 \sin \theta = 6 \cdot \frac{y}{r}$

$r^2 = 6y \Rightarrow x^2 + y^2 - 6y = 0$
 is circle with centre $(0, 3)$

$I = \int_0^\pi \int_{4 \sin \theta}^{6 \sin \theta} (r^3 dr) d\theta$

$I = \int_0^\pi \left(\frac{r^4}{4} \right)_{4 \sin \theta}^{6 \sin \theta} d\theta$

$I = \frac{1}{4} \int_0^\pi [(6 \sin \theta)^4 - (4 \sin \theta)^4] d\theta$

$I = \frac{1}{4} \int_0^\pi (6^4 - 4^4) \sin^4 \theta d\theta$

$$I = \frac{6^4 - 4^4}{4} \int_0^{\pi} \sin^4 \theta d\theta$$

$$= 260 \int_0^{\pi/2} \sin^4 \theta d\theta$$

$$\int_0^{\pi/2} \sin^n \theta d\theta = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \dots \frac{\pi}{2}$$

$$= 260 \cdot 2 \int_0^{\pi/2} \sin^4 \theta d\theta$$

$$= 520 \left(\frac{4-1}{4} \right) \left(\frac{4-3}{2} \right) \cdot \frac{\pi}{2}$$

$$= 520 \left(\frac{3}{4} \right) \left(\frac{1}{2} \right) \cdot \frac{\pi}{2}$$

$$= \frac{520}{16} \cdot \frac{3\pi}{2}$$

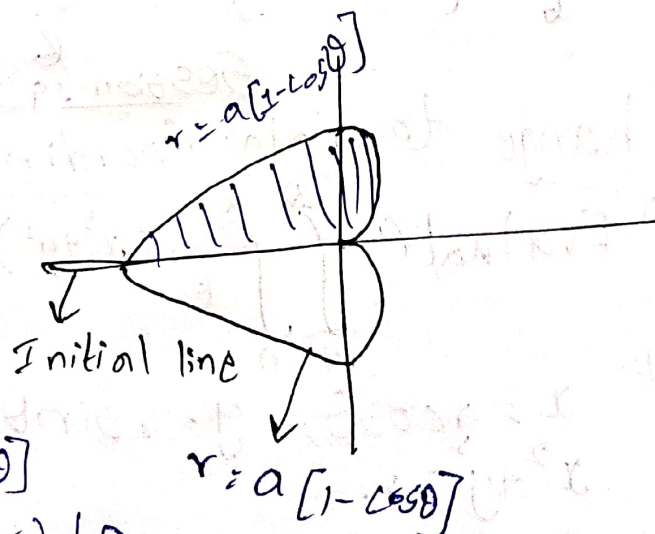
2) Evaluate $\iint r \sin \theta dr d\theta$ over the
 R: Cordiod... $r = a[1 - \cos \theta]$ {above the
 initial line.

$$r: 0 \rightarrow a[1 - \cos \theta]$$

$$\theta: 0 \rightarrow \pi$$

$$I = \int_0^{\pi} \int_0^{a[1-\cos \theta]} r \sin \theta dr d\theta$$

$$I = \int_0^{\pi} \sin \theta \left(\int_0^{a[1-\cos \theta]} r dr \right) d\theta$$



$$= \int_{\theta=0}^{\pi} \sin \theta \cdot \left(\frac{r^2}{2} \right) d\theta$$

$$= \frac{1}{2} \int_{\theta=0}^{\pi} \sin \theta [a(1-\cos \theta)]^2 d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi} \sin \theta (1-\cos \theta)^2 d\theta$$

$$\left(\because \int f(x) [f(x)]^n dx = \frac{f(x)^{n+1}}{n+1} + C \right)$$

$$= \frac{a^2}{2} \left[\frac{(1-\cos \theta)^{2+1}}{2+1} \right]_0^{\pi}$$

$$= \frac{a^2}{2} \left[\frac{(1+1)^{2+1}}{2+1} \right] = \frac{a^2}{2} \cdot \frac{1}{3} \left[(1-\cos \theta)^3 \right]_0^{\pi}$$

$$= \frac{a^2}{6} \left[(1-\cos \pi)^3 - (1-\cos 0) \right]$$

$$= \frac{a^2}{6} [8 - 0] = \frac{a^2}{6} \cdot 8 = \frac{4a^2}{3} //$$

Change to Polar Coordinates

Q) Evaluate $\int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dx dy$

Sol:-

$$x = r \cos \theta, \quad y = r \sin \theta, \quad dx dy = r dr d\theta$$

$$x^2 + y^2 = r^2$$

$$I = \int_0^{\infty} \int_0^{\infty} e^{-r^2} r dr d\theta$$

$$x: 0 \rightarrow \infty$$

$$y: 0 \rightarrow \infty$$

$$r: 0 \rightarrow \infty$$

$$\theta: 0 \rightarrow \frac{\pi}{2}$$

$$I = \int_{\theta=0}^{\pi/2} \left(\int_{r=0}^{\infty} e^{-r^2} r dr \right) d\theta$$

$$r^2 = t$$

$$2r dr = dt$$

$$r dr = \frac{dt}{2}$$

$$\text{L.L. } r=0 \Rightarrow t=0$$

$$\text{U.L. } r=\infty \Rightarrow t \rightarrow \infty$$

$$= \int_0^{\pi/2} \left(\int_{t=0}^{\infty} e^{-t} \frac{dt}{2} \right) d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \left(\frac{e^{-t}}{-1} \right)_0^{\infty} d\theta$$

$$x=0$$

$$-\frac{1}{2} \int_0^{\pi/2} [0 - 1] d\theta$$

$$= +\frac{1}{2} \int_0^{\pi/2} 1 d\theta$$

$$= \frac{1}{2} (0)_0^{\pi/2}$$

$$= \frac{1}{2} \cdot \frac{\pi}{2}$$

$$= \frac{\pi}{4} //$$

18-03-25

② By taking Polar co-ordinates evaluate

$$\int_0^{\sqrt{a^2-y^2}} \int_0^{\sqrt{a^2-y^2}} (a^2+y^2) dx dy$$

$$\text{Sol: } x = r \cos \theta, y = r \sin \theta, dx dy = r dr d\theta$$

$$I = \int_{y=0}^a \int_{x=0}^{\sqrt{a^2-y^2}} (r^2) r dr d\theta$$

Given limits

$$x: 0 \rightarrow \sqrt{a^2 - y^2}$$

$$y: 0 \rightarrow a$$

Given $x=0$ &

$$x = \sqrt{a^2 - y^2}$$

$$x^2 = a^2 - y^2$$

$$x^2 + y^2 = a^2$$

New limits

$$r: 0 \rightarrow a$$

$$\theta: 0 \rightarrow \pi/2$$

$$I = \int_0^{\pi/2} \int_0^a r^3 dr d\theta$$

$$= \int_0^{\pi/2} \left(\frac{r^4}{4} \right)_0^a d\theta$$

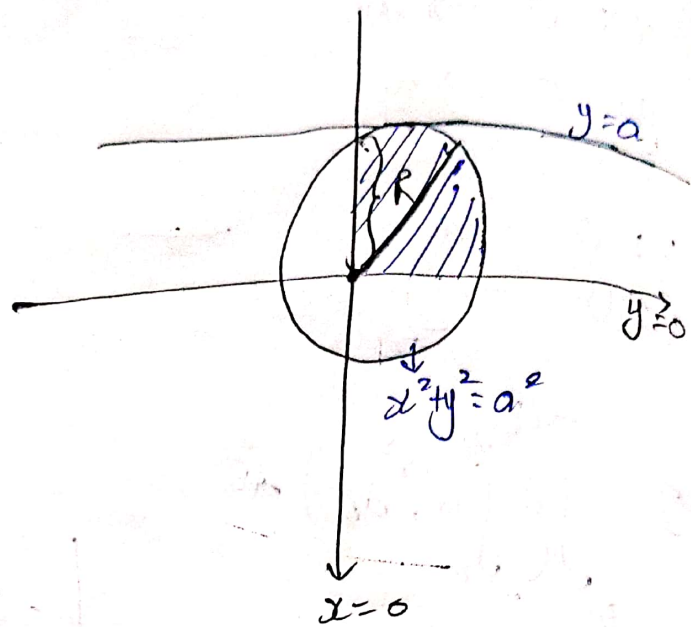
$$= \frac{1}{4} \int_0^{\pi/2} [a^4 - 0] d\theta$$

$$= \frac{1}{4} \int_0^{\pi/2} a^4 d\theta$$

$$= \frac{a^4}{4} \int_0^{\pi/2} 1 d\theta$$

$$= \frac{a^4}{4} \int_0^{\pi/2} \theta$$

$$= \frac{a^4}{4} (\theta)_0^{\pi/2} \Rightarrow \frac{a^4}{4} \cdot \frac{\pi}{2} = \frac{\pi a^4}{8} //$$



② Using polar coordinates evaluate

$$\int_0^a \int_0^{\sqrt{a^2-x^2}} y \sqrt{x^2+y^2} dy dx$$

Sol:- Take $x = r \cos \theta$, $y = r \sin \theta$, $dx dy = r dr d\theta$

$$I = \int_{x=0}^a \int_{y=0}^{\sqrt{a^2-x^2}} y \sqrt{x^2+y^2} dy dx$$

$$I = \int_{x=0}^a \int_{y=0}^{\sqrt{a^2-x^2}} r \sin \theta \sqrt{r^2} r dr d\theta$$

$$I = \int_{x=0}^a \int_{y=0}^{\sqrt{a^2-x^2}} r^3 \sin \theta dr d\theta$$

Given limits

$$y: a \rightarrow \sqrt{a^2-x^2}$$

$$x: 0 \rightarrow a$$

i.e., Squaring on both sides

$$y^2 = a^2 - x^2$$

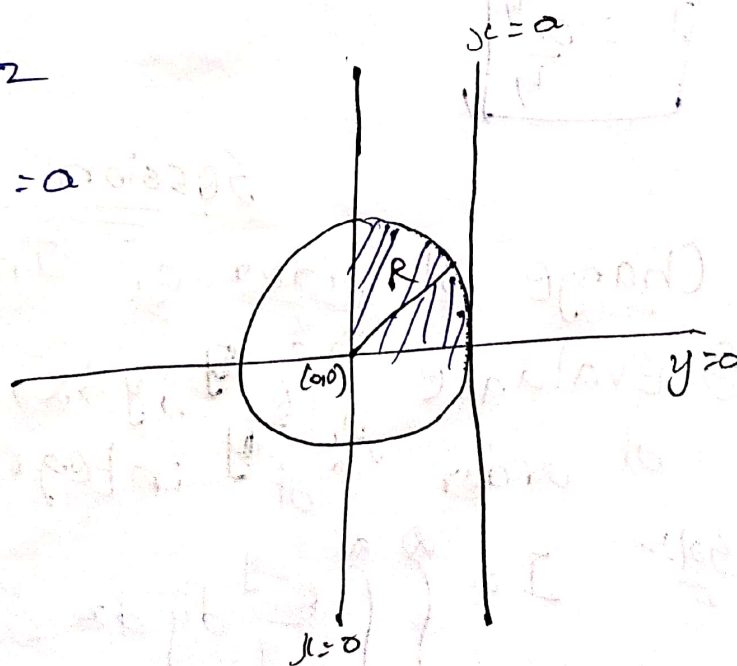
$$y=0; x^2 + y^2 = a^2$$

* For $x=0 \Rightarrow x=a$

~~$$x: 0 \rightarrow \frac{\pi}{2}$$~~

$$r: 0 \rightarrow a$$

$$\theta: 0 \rightarrow \frac{\pi}{2}$$



$$I = \int_{\theta=0}^{\pi/2} \int_{r=0}^{2a} r^3 dr \sin \theta d\theta$$

$$I = \int_0^{\pi/2} \left[\frac{r^4}{4} \right]_0^{2a} \sin \theta d\theta$$

$$I = \frac{1}{4} \int_0^{\pi/2} a^4 \sin \theta d\theta$$

$$I = \frac{a^4}{4} \int_0^{\pi/2} 1 \cdot \sin \theta d\theta$$

$$I = \frac{a^4}{4} [-\cos \theta]_0^{\pi/2}$$

$$I = \frac{a^4}{4} \left[-\cos \frac{\pi}{2} + \cos 0 \right]$$

$$I = \frac{a^4}{4} [0 + 1]$$

$$\boxed{I = \frac{a^4}{4}}$$

Session - 20

Change of order of Integration.

① Evaluate $\int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dy dx$, by using change of order of integration.

Sol:-

$$I = \int_{x=0}^{\infty} \int_{y=x}^{\infty} \frac{e^{-y}}{y} dy dx$$

Given limits

$$y: x \rightarrow \infty$$

$$x: 0 \rightarrow \infty$$

$$y=x \text{ \& } y \rightarrow \infty$$

$$x=0 \text{ \& } x \rightarrow \infty$$

New limits

$$x: 0 \rightarrow y$$

$$y: 0 \rightarrow \infty$$

$$I = \int_{y=0}^{\infty} \left[\int_{x=0}^y \frac{e^{-y}}{y} dx \right] dy$$

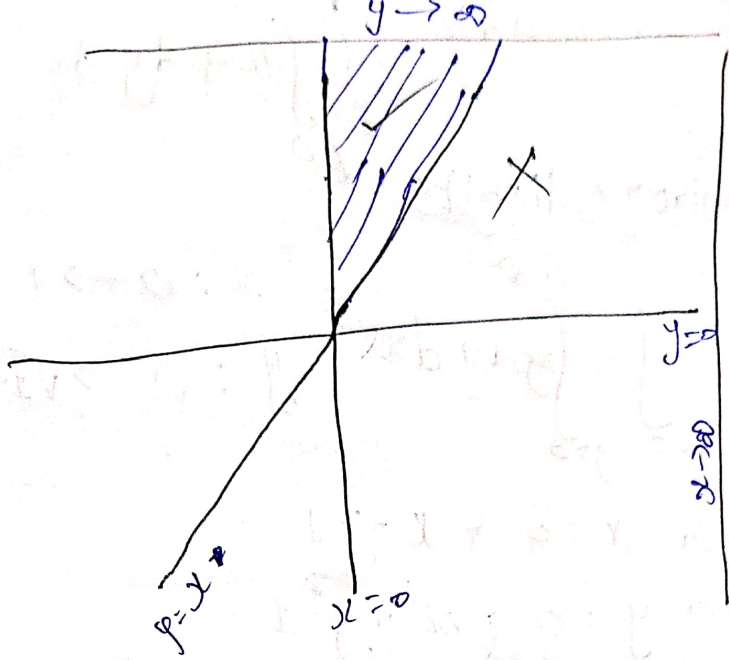
$$= \int_{y=0}^{\infty} \left[\frac{e^{-y}}{y} \int_{x=0}^y 1 dx \right] dy$$

$$= \int_{y=0}^{\infty} \frac{e^{-y}}{y} (x)_0^y dy$$

$$= \int_{y=0}^{\infty} \frac{e^{-y}}{y} \times y dy$$

$$= \int_0^{\infty} \frac{e^{-y}}{-1} \Rightarrow \left[\frac{e^{-y}}{-1} \right]_0^{\infty}$$

$$= -[0 - 1] = 1 //$$



② Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} xy \, dy \, dx$

Sol:-

Given limits

$$\int_{x=0}^1 \int_{y=0}^{\sqrt{1-x^2}} xy \, dy \, dx \quad \begin{array}{l} x: 0 \rightarrow 1 \\ y: 0 \rightarrow \sqrt{1-x^2} \end{array}$$

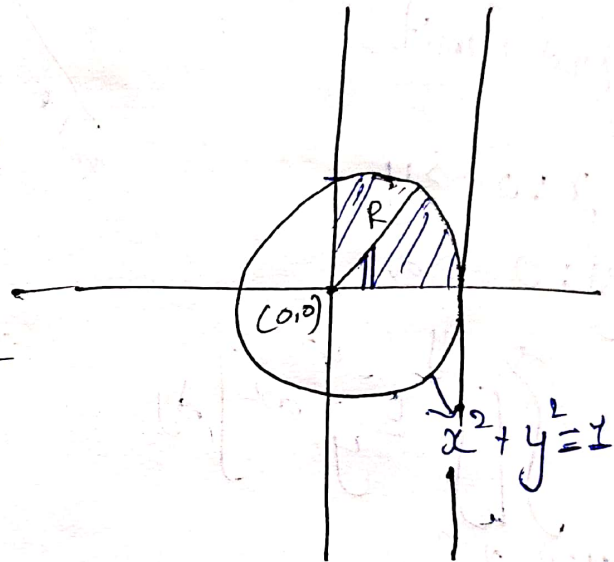
For $x=0$ & $x \rightarrow 1$

for $y=0$ & $x^2+y^2=1$

New limits

$$x: 0 \rightarrow \sqrt{1-y^2}$$

$$y: 0 \rightarrow 1$$



$$I = \int_{y=0}^1 \left[\int_{x=0}^{\sqrt{1-y^2}} xy \, dx \right] dy$$

$$I = \int_{y=0}^1 y \left(\frac{x^2}{2} \right)_0^{\sqrt{1-y^2}} dy$$

$$I = \frac{1}{2} \int_0^1 y (1-y^2) dy$$

$$I = \frac{1}{2} \int_0^1 (y - y^3) dy$$

$$I = \frac{1}{2} \left[\frac{y^2}{2} - \frac{y^4}{4} \right]_0^1$$

$$\frac{1}{2} \left[\frac{1}{2} - \frac{1}{4} \right]$$

$$\frac{1}{2} \left[\frac{2-1}{4} \right]$$

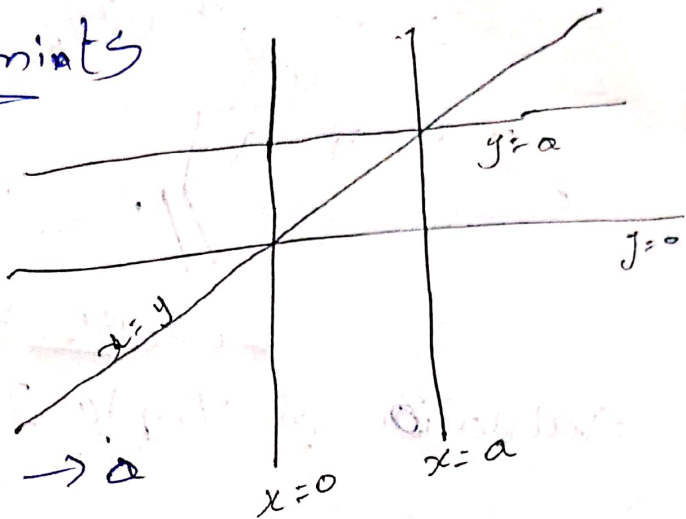
$$\frac{1}{2} \left[\frac{1}{4} \right] = \frac{1}{8}$$

③ Evaluate $\int_0^a \int_y^a \frac{4x}{x^2+y^2} dx dy$

Sol:- Given limits

$$x: y \rightarrow a$$

$$y: 0 \rightarrow a$$



New limits

$$x: 0 \rightarrow a$$

$$y: 0 \rightarrow x$$

$$I = \int_{x=0}^a \left[\int_{y=0}^x \frac{4x}{x^2+y^2} dy \right] dx$$

$$I = 4 \int_0^a \left[\frac{x}{x^2+y^2} dy \right]_{y=0}^x dx$$

$$I = 4 \int_0^a x \left(\frac{1}{x^2+y^2} dy \right) dx$$

$$I = 4 \int_0^a x \cdot \frac{1}{x} \left[\tan^{-1}\left(\frac{y}{x}\right) \right]_{y=0}^x dx$$

$$I = 4 \int_0^a \left[\tan^{-1}\left(\frac{x}{x}\right) - (\tan^{-1}(0)) \right] dx$$

$$I = 4 \int_0^{\frac{\pi}{4}} \left(\frac{\pi}{4} - 0 \right) dx$$

$$= \frac{\pi}{4} \cdot 4 (x)_0^{\frac{\pi}{4}}$$

$$\boxed{I = \pi a}$$

Session-21

19/03/25

Evaluation of Triple Integral.

$$(1) \int_{x=0}^1 \int_{y=0}^2 \int_{z=0}^3 xy^2z \, dx \, dy \, dz$$

$$\text{Sol:-} \int_{x=0}^1 \int_{y=0}^2 (xy^2) \left(\frac{z^2}{2} \right)_0^3 dx \, dy$$

$$= \int_{x=0}^1 \int_{y=0}^2 (xy^2) \left(\frac{9}{2} - \frac{0}{2} \right) dx \, dy \Rightarrow \frac{9}{2} \int_{x=0}^1 \int_{y=0}^2 (xy^2) dy \, dx$$

$$= \frac{9}{2} \int_{x=0}^1 x \left[\frac{y^3}{3} \right]_0^2 dx \Rightarrow \frac{9}{2} \int_{x=0}^1 x \left[\frac{8}{3} - \frac{0}{3} \right] dx$$

$$= \frac{9 \times 8}{2 \times 3} \int_{x=0}^1 x \, dx$$

$$= 12 \left[\frac{x^2}{2} \right]_0^1 \Rightarrow 12 \left[\frac{1}{2} - 0 \right] = 6 //$$

$$2) \int_0^a \int_0^b \int_0^c xy^2 \cdot dx \cdot dy \cdot dz$$

Sol:- $I = \int_{z=0}^a \int_{y=0}^b \left[y^2 \left[\frac{x^2}{2} \right]_0^c \right] dy dz$

$$= \int_{z=0}^a \int_{y=0}^b y^2 \left[\frac{c^2}{2} \right] dy dz = \frac{c^2}{2} \int_{z=0}^a \left[\int_{y=0}^b (y^2 \cdot dy) \right] dz$$

$$= \frac{c^2}{2} \int_{z=0}^a \left[\frac{y^3}{3} \right]_0^b dz = \frac{c^2}{2} \int_0^a \left[\frac{b^3}{3} - 0 \right] dz$$

$$= \frac{c^2}{2} \times \frac{b^3}{3} \int_0^a 1 \cdot dz$$

$$= \frac{c^2 b^3}{6} [z]_0^a$$

$$= \frac{c^2 b^3}{6} \times a = \frac{a b^3 c^2}{6} //$$

$$3) \int_0^1 \int_0^x \int_0^{x+y} xyz \cdot dx \cdot dy \cdot dz$$

Sol:- $I = \int_{x=0}^1 \int_{y=0}^x \left[\frac{z^2}{2} \right]_0^{x+y} dy dx$

$$\int_0^1 \int_0^x xy \left[\frac{(x+y)^2}{2} - 0 \right] dy dx$$

$$\frac{1}{2} \int_0^1 \int_0^x [xy(x^2 + y^2 + 2xy)] dy dx$$

$$\frac{1}{2} \int_0^1 \int_0^x [x^3y + xy^3 + 2x^2y^2] dy dx$$

$$\frac{1}{2} \int_0^1 \left[\frac{x^3y^2}{2} + \frac{xy^4}{4} + 2x^2 \frac{y^3}{3} \right]_0^x dx$$

$$\frac{1}{2} \int_0^1 \left[\frac{x^3 \cdot x^2}{2} + \frac{x \cdot x^4}{4} + \frac{2x^2 \cdot x^3}{3} \right] dx$$

$$\frac{1}{2} \int_0^1 \left[\frac{6x^5 + x^5 + 8x^5}{12} \right] dx$$

$$\frac{1}{2} \int_0^1 \left[\frac{14x^5}{12} \right] dx$$

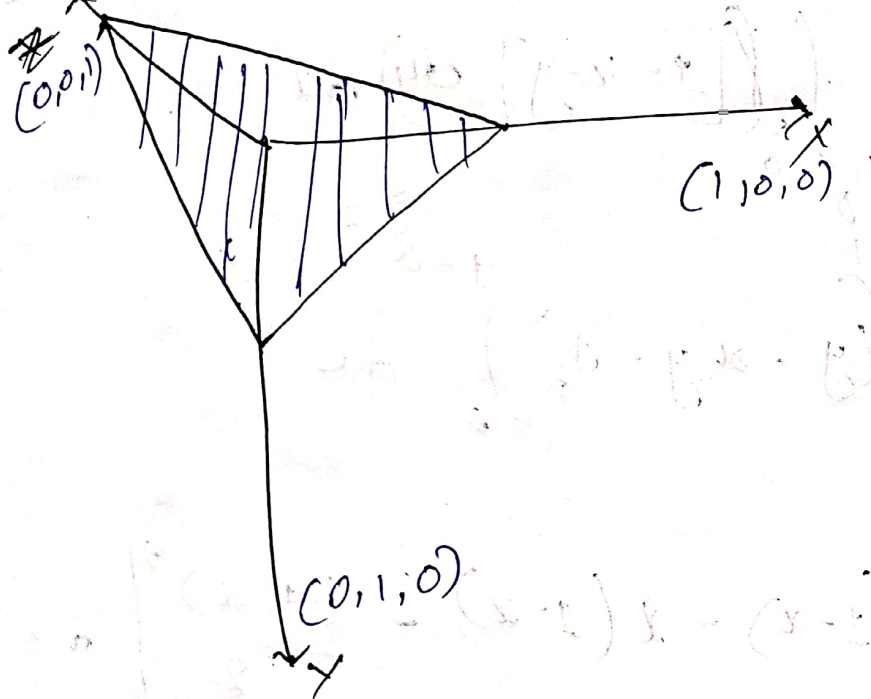
$$\frac{1}{2} \cdot \frac{14}{12} \left(\frac{x^6}{6} \right)_0^1$$

$$\frac{1}{2} \cdot \frac{14}{12} \left(\frac{1}{6} \right) = \frac{14}{144}$$

Application of Triple Integral

① Volume:- Calculate the volume of the solid bounded by the planes $x=0, y=0$ and $z=0$ and $x+y+z=1$.

$$V = \iiint dx dy dz \quad (\text{or}) \quad \iiint dV$$



For z limits

$$z: 0 \rightarrow 1-x-y \quad \text{from } x+y+z=1 \Rightarrow z=1-x-y$$

For y limits: Take $z=0$ in $x+y+z=1$
 $x+y=1$

$$x = 1-y$$
$$y = 1-x$$

For x limits

Take $y=0$

$$x+y=1$$

$$x=1 \Rightarrow x: 0 \rightarrow 1$$

$$I = \int_{x=0}^1 \int_{y=0}^{1-x} \int_{z=0}^{1-x-y} 1 \, dz \, dy \, dx$$

$$= \int_0^1 \int_0^{1-x} [z]_0^{1-x-y} dy \, dx$$

$$= \int_0^1 \left(\int_0^{1-x} [1-x-y] dy \right) dx$$

$$= \int_0^1 \left(y - xy - \frac{y^2}{2} \right)_0^{1-x} dx$$

$$= \int_0^1 \left[(1-x) - x(1-x) - \frac{(1-x)^2}{2} \right] dx$$

$$= \int_0^1 \left[(1-x)(1-x) - \frac{(1+x^2-2x)}{2} \right] dx$$

$$= \int_0^1 \left[(1-x)^2 - \frac{(1-x)^2}{2} \right] dx$$

$$= \int_0^1 (1-x)^2 \left[1 - \frac{1}{2} \right] dx = \frac{1}{2} \int_0^1 (1-x)^2 dx$$

$$\begin{aligned}
 &= \frac{1}{2} \left[x - \frac{x^3}{3} \right]_0^1 = \frac{1}{2} \left[\frac{(1-x)^3}{3} \right]_0^1 \\
 &= \frac{1}{2} \left[1 - \frac{1}{3} \right] = \frac{1}{2} \left[\frac{2}{3} \right] = \frac{1}{3} \\
 &= \frac{1}{2} \left[\frac{2}{3} \right] = \frac{1}{3}
 \end{aligned}$$

Q. Volume bounded by $2x + y + z = 6$ that lies in Ist octant

Sol: $V = \iiint dxdydz$

$$2x + y + z = 6$$

$$z = 6 - 2x - y$$

for z limits
 $z: 0 \rightarrow 6 - 2x - y$

for y limits
 Take $z = 0$

$$2x + y = 6$$

$$y = 6 - 2x$$

$y: 0 \rightarrow 6 - 2x$
 $2x = 6 - y \Rightarrow x = \frac{6-y}{2}$

for x limits

Take $y = 0$

$$x = \frac{6-0}{2}$$

$$x = 3$$

$$x: 0 \rightarrow \frac{6}{2}$$



$$V = \int_{x=0}^{\frac{3}{2}} \int_{y=0}^{3-2x} \int_{z=0}^{3-2x-y} (1 \cdot dz) \cdot dy \cdot dx$$

$$V = \int_{x=0}^{\frac{3}{2}} \int_{y=0}^{3-2x} [z]_0^{3-2x-y} dy \cdot dx$$

$$V = \int_{x=0}^{\frac{3}{2}} \int_{y=0}^{3-2x} [3-2x-y] dy \cdot dx$$

$$V = \int_{x=0}^{\frac{3}{2}} \left((3-2x)y - \frac{y^2}{2} \right)_0^{3-2x} dx$$

$$V = \int_{x=0}^{\frac{3}{2}} \left[(3-2x)y(3-2x) - \frac{(3-2x)^2}{2} \right] dx$$

$$V = \int_0^{\frac{3}{2}} \left[(3-2x)^2 - \frac{(3-2x)^2}{2} \right] dx$$

$$V = \int_0^{\frac{3}{2}} \left((3-2x)^2 \left(1 - \frac{1}{2} \right) \right) dx = \frac{1}{2} \int_0^{\frac{3}{2}} (3-2x)^2 dx$$

$$= \frac{1}{2} \left[\frac{(3-x)^3}{-3} \right]_{x=0}^{-2}$$

$$= \frac{1}{2} \left[\frac{(3 - \cancel{2}(\frac{3}{2}))^3}{-3} - (3-0)^3 \right]$$

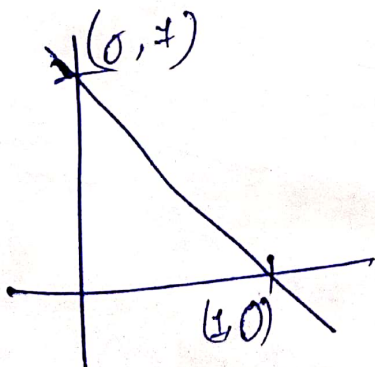
$$= \frac{1}{2} \left[\frac{(3-3)^3}{-3} - (3-0)^3 \right]$$

$$= -\frac{1}{2} [-27]$$

$$= \frac{27}{2} // = \frac{9}{4} //$$

$$\therefore \int (ax+b)^n = \left[\frac{(ax+b)^{n+1}}{\frac{n+1}{a}} \right]$$

Evaluate $\iint dy dx dy$ over 1st quadrant at $x+y=1$.



$$y: 0 \rightarrow 1-x$$

$$x: 0 \rightarrow 1-y$$